



FACULTY OF INFORMATION
TECHNOLOGY

MIDTREM EXAM 1B ACADEMIC YEAR 2025 - 2026

Major/Class : IT/B01
 Course : Application development of advanced web
 Course code : ITE1278E No. of Credits:.....
 Date : 05/01/2026
 Time limit : 120' Exam methods: Practical ...
 Exam code : 001

OPEN EXAM: YES ☒ NO ☐

I. GENERAL REQUIREMENTS:

Build an ASP.NET Core MVC 8.0 web application for product management with 2 sections: Admin (administration) and Client (user). **AI Tools: Allowed for the first 90 minutes**

II. DATABASE (1.5 points)

3 tables:

- Categories: CategoryId, **CategoryName**, **Description**
- Products: ProductId, **ProductName**, **Price**, **Image**, Description, **CategoryId**, IsActive, CreatedDate
- Orders: OrderId, **CustomerName**, **Phone**, **Address**, **ProductId**, **Quantity**, TotalAmount, **OrderDate**, Status (each order contains only **ONE** product)

Requirements: Code First with sample data (3 categories, 10 products)

II. ADMIN (4.0 points)

1. Category Management (1.0 points) (**WEB API**)

- Full CRUD operations (list, create, edit, delete)
- Validation (bold attributes are required (not null). Numeric values must be > 0)

2. Product Management (2.0 points)

- List with pagination (10/page)
- Search by ProductName and filter by category
- Full CRUD operations (list, create, edit, delete)
- Image upload, validation (bold attributes are required (not null). Numeric values must be > 0)

3. Order Management (1.0 points)

- Order list
- View details, update status
- Statistics: total orders, total revenue

III. CLIENT (3.5 points)

1. Home Page (1.5 points)

- Product list (IsActive=true)
- Pagination (12 product/page)
- Category menu

2. Product Details (1.0 points)

- Full information

- Order form on page

3. Place Order (1.0 points)

- Form: name, phone, address, quantity
- Validation, auto-calculate total amount
- Save with Status="Pending", success notification (PAGE)

IV. UI & TECHNICAL (1.0 points)

- Bootstrap 5 responsive (0.5 points)
- Separate layouts for Admin/Client (0.5 points)

BONUS POINTS (max:1.0 points)

- Admin Authentication (0.5 points)
- Statistics Chart (0.5 points)

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No explanation should be carried out